3.8 Comparing and Performing Operations With Absolute Value

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<tr><td>Benchmark</td><td colspan="2">MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.</td><td>Learning Targets</td><td>I can compare the absolute value of two numbers.
I can perform operations that involve the absolute values of two numbers.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="2">MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.</td></tr>
<tr><td>Essential Questions</td><td colspan="4">How can inequality symbols be used to compare absolute values?
How can number lines be used to compare absolute values?
How do you perform operations with absolute values?
How do you compare absolute values in real-world contexts?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">Students interpret and compare absolute value in real-world contexts. Lesson activities leverage strategies and tools like number lines and inequality symbols to represent absolute values in context in addition to using the order of operations to evaluate expressions that include absolute values.</td></tr>
<tr><td rowspan="4">Component Summary</td><td>3.8.1 Warm-Up (~10 min.)</td><td>Discuss real-world applications of concepts learned so far in the unit (SE p. 117)</td><td>3.8.4 Collaboration (5-10 min.)</td><td>Collaboratively compare values and absolute values (SE p. 119)</td></tr>
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<tr><td>3.8.3 Guided Instruction (10 min.)</td><td>Learn how to compare absolute values and consider differences with comparing values (SE p. 118)</td><td>3.8.6 Your Turn! (5 min.)</td><td>Practice comparing values and absolute values and evaluating expressions that include absolute values (SE p. 121)</td></tr>
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<tr><td colspan="2"><u>Key Terms:</u> N/A

<u>Additional Terms:</u> N/A</td><td>(Optional) Pictures or video clips of real-world contexts</td><td>(Optional) Prepare an anchor chart with examples and visual models of key vocabulary and symbols.</td></tr>
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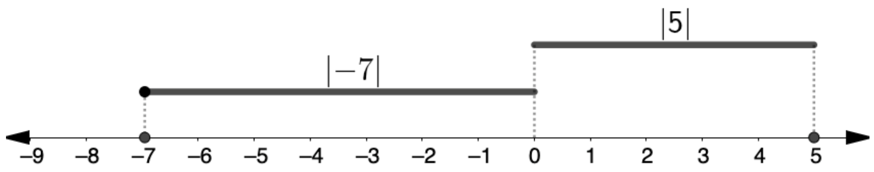
Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle using the correct vocabulary when reading the mathematical statements in 3.8.3 Guided Instruction. • Students may not understand the difference between comparing values and comparing absolute values. • Students may struggle interpreting the meanings of the words elevation, altitude, and sea level.	• Model how to read the mathematical statements in 3.8.3 Guided Instruction. • Circulate and check students' comparisons for accuracy during 3.8.4 Collaboration. • Consider creating and displaying an anchor chart with examples and visual models of key vocabulary and symbols.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Take Turns In 3.8.4 Collaboration, students take turns creating mathematical comparison statements and writing the corresponding written statements. Prompt students to explain how they know their statements are true.	**MA.K12.MTR.6.1: Reasonableness** In 3.8.6 Your Turn!, students compare and evaluate expressions with absolute values. Students consider the reasonableness of their solutions in the given context.

Differentiate Instruction

Number lines offer spatial and visual strategies for students making sense of absolute value of numbers as their distance from zero. Note the example pictured here from 3.8.5 Guided Instruction. Encourage students to utilize number lines throughout this lesson either to make sense of the numbers or verify their answers.



Lesson Notes:

3.8.1 Warm-Up: Rational Numbers in the Real World

Component Overview: *Students discuss real-world applications of concepts learned so far in the unit.*

3.8.1 Warm-Up: Rational Numbers in the Real World

For each of the following problems, ask a classmate for his/her explanation. Once you have received their explanation, write a summary in the space provided in the box. Your classmate should read your summary and initial that it is correct.
Repeat this with a different classmate for each of the remaining problems.

<table>
<tr>
<td>1. Describe the relationship between two numbers that have the same absolute value.
Answers will vary. They are opposites. They have the same distance from zero on a number line.
Initials: ____</td>
<td>2. In an American football game, the offensive team makes gains or losses on each play. Gains are represented by positive values and losses by negative values. What does 0 represent in this context?
Answers will vary. Zero represents a play where no gains or losses are made.
Initials: ____</td>
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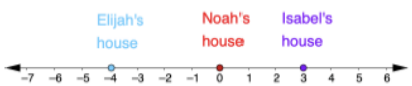
<td colspan="2">3. Elijah, Noah, and Isabel live on the same street. The locations of their houses are modeled on the number line.


Describe where they live relative to each other in two sentences. Include the words distance and absolute value.
Answers will vary. Elijah's house is a distance of 4 blocks from Noah's house and Isabel's is a distance of 3 blocks. The absolute value of the location of Elijah's house is greater than the absolute value of the location of Isabel's.
Initials: ____</td>
</tr>
</table>

Model active listening techniques such as revoicing for students. Remind students to record summaries of their partners' explanations. For example, "So what I am hearing you say is…Is that correct?"

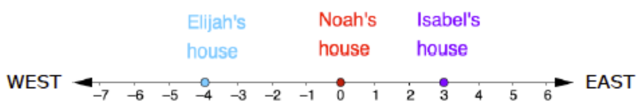
3.8.2 Exploration: Comparing and Interpreting Absolute Value in Context

Component Overview: *Students interpret and compare absolute values in contexts. Students use a number line to aid in understanding their comparisons. The context explored involves locations east and west of a house.*

3.8.2 Exploration: Comparing and Interpreting Absolute Value in Context

1. Recall this scenario from the Warm-Up with some additional information.

   Elijah, Noah, and Isabella live on the same street that runs west to east. The locations of their houses are modeled on the number line, where each unit represents 1 kilometer.



   a. Noah's house is located at 0 on the number line model. Represent the distance from Noah's house to both Isabella's house and Elijah's house using absolute value.

	Elijah's House	Isabella's House
Distance from Noah's House	$\lvert -4 \rvert$	$\lvert 3 \rvert$

   b. How do the absolute values in the table compare to each other?

   $\lvert -4 \rvert > \lvert 3 \rvert$

   c. Isabella lives 3 kilometers to the (○ east ○ west) of Noah. Elijah lives 4 kilometers to the (○ east ○ west) of Noah. Isabella lives (○ closer to ○ farther from) Noah than Elijah does.

During instruction, note the advantages and disadvantages of using horizontal and vertical number lines in different contexts.

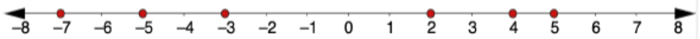
- *What does zero represent in the context?*
- *What represents positive values?*
- *What represents negative values?*

3.8.3 Guided Instruction: Comparing Absolute Values

Component Overview: *Students learn how to compare values and absolute values. Students use a number line to evaluate mathematical comparison statements as true or false. Students discuss and write justifications for their answers.*

3.8.3 Guided Instruction: Comparing Absolute Values

A number line is shown with the integers $-7, -5, -3, 2, 4,$ and 5 plotted.



1. Discuss with your partner which statements are true. Be prepared to explain your thinking.

Mathematical Statement	Explanation
$-7 < -3$	*True. -7 is farther to the left on the number line than -3.*
$\lvert -7 \rvert$ is 7	*True. -7 is 7 units from zero, so the absolute value of -7 is 7.*
$\lvert -3 \rvert$ is 3	*True. -3 is 3 units from zero, so the absolute value of -3 is 3.*
$\lvert -7 \rvert < \lvert -3 \rvert$	*False. -7 is 7 units from zero while -3 is only 3 units from zero, so $\lvert -7 \rvert$ is greater than $\lvert -3 \rvert$.*

2. Does the comparison of -7 and -3 match the comparison of $\lvert -7 \rvert$ and $\lvert -3 \rvert$?
   No. $-7 < -3$, but $\lvert -7 \rvert > \lvert -3 \rvert$.

3. With your partner, discuss why comparing two negative values and comparing the absolute value of the same two negative numbers may be different. Summarize your discussion below.
   Answers will vary. Comparing the values of two negative numbers is a comparison of each number's worth, and comparing the absolute values of two negative numbers is a comparison of their distance from zero. For two negative numbers, the number that is farther from zero on the number line has a greater absolute value, but a lower value.

Before students begin the questions, model how to read a few of the mathematical statements. At the end, debrief the key concepts with a discussion to provide an auditory entry point and check for student understanding.

3.8.3 Guided Instruction: Comparing Absolute Values (continued)

4. For each row, use the symbols $>, <$, or $=$ to compare each pair of values.

	Inequality	Absolute Value Inequality
Pair 1	$4 > -5$	$\lvert 4 \rvert < \lvert -5 \rvert$
Pair 2	$-3 < 2$	$\lvert -3 \rvert > \lvert 2 \rvert$
Pair 3	$6 > -6$	$\lvert 6 \rvert = \lvert -6 \rvert$

5. What do you notice about the comparisons?
Answers will vary. The comparisons of the values of each pair of numbers use different symbols than the comparisons of their absolute values.

6. When comparing the absolute values of two numbers, numbers that are closer to zero will always have the *smaller* absolute value.

Consider creating and displaying an anchor chart for students to reference when reading and writing symbols.

Symbol	Meaning in Words
$<$	"is less than"
$>$	"is greater than"
$=$	"is" or "is equal to"
$\lvert x \rvert$	"the absolute value of x"

Note that "<" could be interpreted as greater or less than depending on the way students read the inequality statement.

For example: $-2 > -7$ could be read as "negative seven is less than negative two" OR "negative two is greater than negative seven."

3.8.4 Collaboration: Inequality Mix and Match

Component Overview: *Students create mathematical statements by selecting two expressions and a comparison symbol from a table. Then, the student's partner represents the mathematical statement with a written mathematical statement. Finally, students switch roles and repeat the activities.*

3.8.4 Collaboration: Inequality Mix and Match

Work with your partner to complete the table.

Partner A will select two numbers and one comparison symbol. At least one of the numbers must be written as an absolute value.

Partner B will then write the mathematical statement in words. The following phrases may be helpful: is equal to, the absolute value of, is greater than, is less than. Switch roles until each partner has taken two turns. An example is shown in the first row.

Numbers					Symbol
-0.7	$\lvert -9 \rvert$	11	14	$\lvert -8 \rvert$	$<$
$\frac{-6}{2}$	-2.2	$\lvert 2.5 \rvert$	8	$\lvert 0.7 \rvert$	$=$
$\lvert -14 \rvert$	$\lvert 20 \rvert$	$\frac{7}{2}$	$\lvert 15 \rvert$	$\left\lvert -\frac{5}{2} \right\rvert$	$>$

Partner	Mathematical Statement	Statement in Words
Example	$\lvert 3 \rvert < 4$	*the absolute value of 3 is less than 4*
Partner A	*Answers will vary.*	*Answers will vary.*
Partner B	*Answers will vary.*	*Answers will vary.*
Partner A	*Answers will vary.*	*Answers will vary.*
Partner B	*Answers will vary.*	*Answers will vary.*

Take Turns
This activity provides students opportunities to take turns creating mathematical comparison statements. To ensure the activity is productive:

- *Point students toward the example shown in the workbook.*
- *Check students' statements for accuracy.*
- *Facilitate discussions to address misconceptions as needed.*

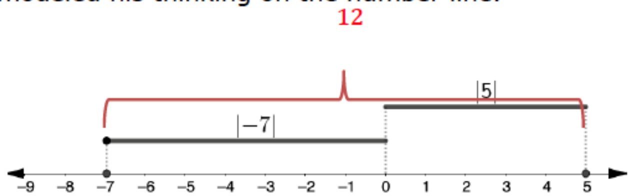
Challenge students who finish early to create real-world examples that can be represented by their mathematical statements.

3.8.5 Guided Instruction: Operations Using Absolute Values

Component Overview: *Students learn how to perform operations with absolute values.*

3.8.5 Guided Instruction: Operations using Absolute Values

1. Ty'shaun and Gabby are trying to determine the sum of $|-7|$ and $|5|$ and explained their strategies. Ty'shaun modeled his thinking on the number line.



   Gabby's explanation was "I know $|-7|$ is a distance of 7 from 0 and $|5|$ is a distance of 5 from 0. The sum of 7 and 5 is 12, so the sum of $|-7|$ and $|5|$ is also 12."
   Explain which strategy makes more sense to you.
   Answers will vary. Gabby's strategy makes sense to me because she just adds the absolute values. Ty'shaun's strategy makes sense to me because he modeled addition on the number line to find the total distance.

2. Find the value of each expression.

   $|-18| - |-13|$
   5

   $12 \times |-4|$
   48

   $\dfrac{|45|}{|-3|}$
   15

   $|-22| + |31| + |-40|$
   93

Consider modeling how to create and evaluate an absolute value expression. Use the Think Aloud technique to help students understand how you are using the order of operations. Check for understanding before students work independently.

Absolute values are grouping symbols, so they should be evaluated prior to performing other operations. Students would have learned that grouping symbols are evaluated first in elementary school. Avoid referring to PEMDAS or any other acronym. Refer to Unit 13 to preview how the order of operations will be addressed there.

How are Gabby and Ty'shaun's explanations similar? How are they different?

When do we evaluate absolute values when performing operations?

3.8.6 Your Turn!

Component Overview: *Students practice comparing values and absolute values and evaluate absolute value expressions.*

3.8.6 Your Turn!

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

   a. $|-42| > -25$
   b. $-21 < |18|$
   c. $|-582| < |-790|$
   d. $|-512| > 284$
   e. $|-176| = |176|$
   f. $|-38| > 0$

2. An octopus is 47 feet below sea level at the same time a dolphin is at 108 feet below sea level.
   a. Write a value to describe the position of each sea animal relative to sea level.

Octopus	Dolphin
$-47\ feet$	$-108\ feet$

   b. Use an inequality symbol, < or >, to write an inequality comparing the distance each sea animal is from sea level.

   $|-47| < |-108|$ *OR* $|-108| > |-47|$

   c. What does this inequality statement indicate about the locations of each sea animal?
   Answers will vary. The dolphin is farther from the surface of the water than the octopus.

3. Find the value of each expression.
   a. $|-58| + 12 = 70$

   b. $|-13| \times 5 = 65$

   c. $\frac{|-36|}{|9|} = 4$

Prompt students to sketch a number line to represent the situation in question 2 and make meaning in context, if needed.

Students may forget to determine the absolute value before performing the operations in question 3. As you monitor student progress, facilitate a discussion (as needed) about evaluating grouping symbols first.

If students finish early, prompt them to write real-world problems involving comparing absolute values. Ask students to switch problems and compare the absolute values in each other's problems.

3.8.7 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their mastery of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.*

3.8.7 Wrap-Up: Ticket Out the Door

1. The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Vpadina Kaundy, Kazakhstan	$^{-}132$
Laguna del Carbón, Argentina	$^{-}105$
Zuidplaspolder, Netherlands	$^{-}7$
Fort Myers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands Fort Myers, Florida	$\lvert -7 \rvert > \lvert 2 \rvert$

Britton Hill, Lakewood, Florida Laguna del Carbón, Argentina	$\lvert 105 \rvert = \lvert -105 \rvert$

Laguna del Carbón, Argentina Vpadina Kaundy, Kazakhstan	$\lvert -105 \rvert < \lvert -132 \rvert$

Consider a project-based learning experience for students as homework, extending the 3.8.7 Wrap-Up.

- *Map elements of your bedroom, with the level of your pillow or the door handle as "sea level" in the context of elevation.*
- *Provide the "sea level" of at least five objects in your room.*
- *Students can use different measuring tools as long as a key is provided.*